

ARISE Software Analysis & AI Integration Report

Karnataka State Police Intelligence Platform (ARISE)

1. Executive Summary

ARISE is a state-of-the-art, dark-themed intelligence and investigative platform built for the Karnataka State Police. The application features a React-based frontend using modern visualization frameworks (D3.js, Leaflet, and Recharts) coupled with a serverless backend hosted on Zoho Catalyst.

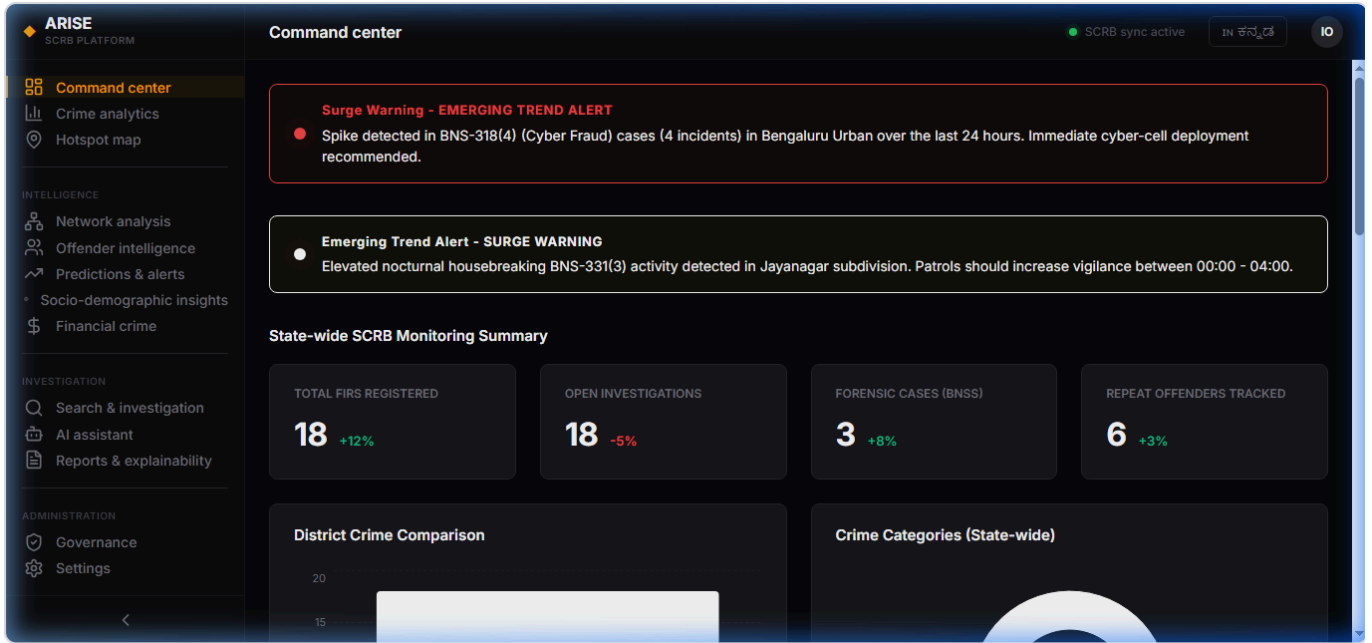
AI-driven capabilities (powered by Zoho Zia Services, QuickML, and Catalyst GLM) are woven deeply into every facet of the platform. Key focus areas include Named Entity Recognition (NER) on FIR narratives, face analytics verification, predictive recidivism scoring, spatial-temporal hotspot anomaly detection, economic vulnerability mapping, and automated, legally compliant (BSA Sec. 63) report generation.

2. Comprehensive Feature Breakdown

Section 1: Command Center (`CommandCenter.jsx`)

The Command Center serves as the central operations room, providing a high-level operational overview of active crimes, alert status, and resource allocation.

- Core Features:**
 - Real-time operational metrics (Total Cases, Active Alerts, Dispatched Units, Open Predictions).
 - Live emergency alerts ticker highlighting high-risk and pending items.
 - BNS Section distribution and regional statistics (Bengaluru, Mysuru, Belagavi, etc.).
 - AI Resource Recommendations Panel:** Suggests strategic deployment of police patrols/units based on real-time spatial predictions and high-risk hotspots.
- Visualizations Used:**
 - Recharts Bar Chart:** Displays the distribution of cases across key BNS (Bharatiya Nyaya Sanhita) sections (e.g., BNS-303, BNS-309(4)).
 - Recharts Pie Chart / Donut Chart:** Renders the regional crime load share by district.
- AI Integration:**
 - Dynamic resource allocation recommendation engine suggesting optimal deployment locations and times.

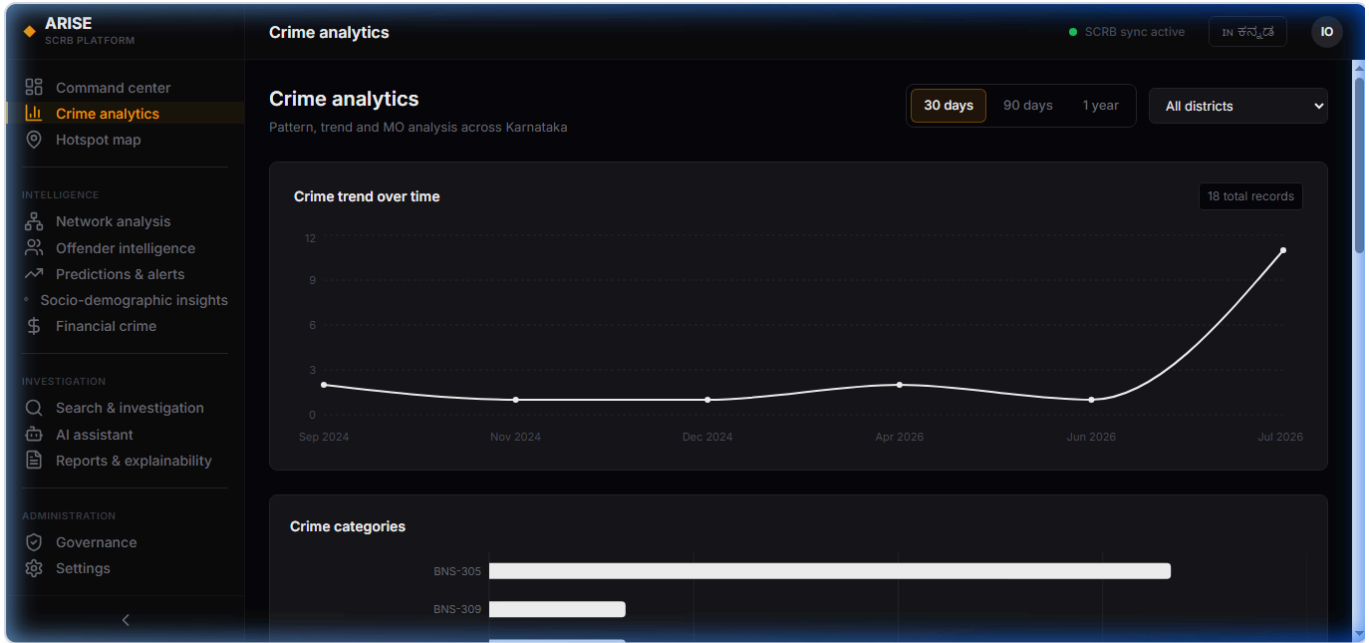


Command Center Screenshot

Section 2: Crime Analytics (`CrimeAnalytics.jsx`)

This dashboard focuses on long-term trends, Modus Operandi (MO) signatures, and predictive temporal distributions.

- **Core Features:**
 - BNS category crime trend comparisons across selected time ranges.
 - Modus Operandi (MO) Profile Analysis, tracking operational signatures such as lock-picking methods, weapon types, and gang affiliations.
 - Spatial risk distribution matrix and temporal peak-time analytics.
- **Visualizations Used:**
 - **Recharts Line Chart:** Multi-series crime rate progression over time.
 - **Recharts Bar Chart:** Breakdown of entry methods, weapons used, and gang involvements.
 - **Custom Risk Matrix Map / Heatmap Grid:** Hourly vs. Day-of-week risk density grids, visually highlighting critical windows (e.g., peak-risk hours on weekends).
- **AI Integration:**
 - MO profile association engine linking crime patterns to specific offender signatures.

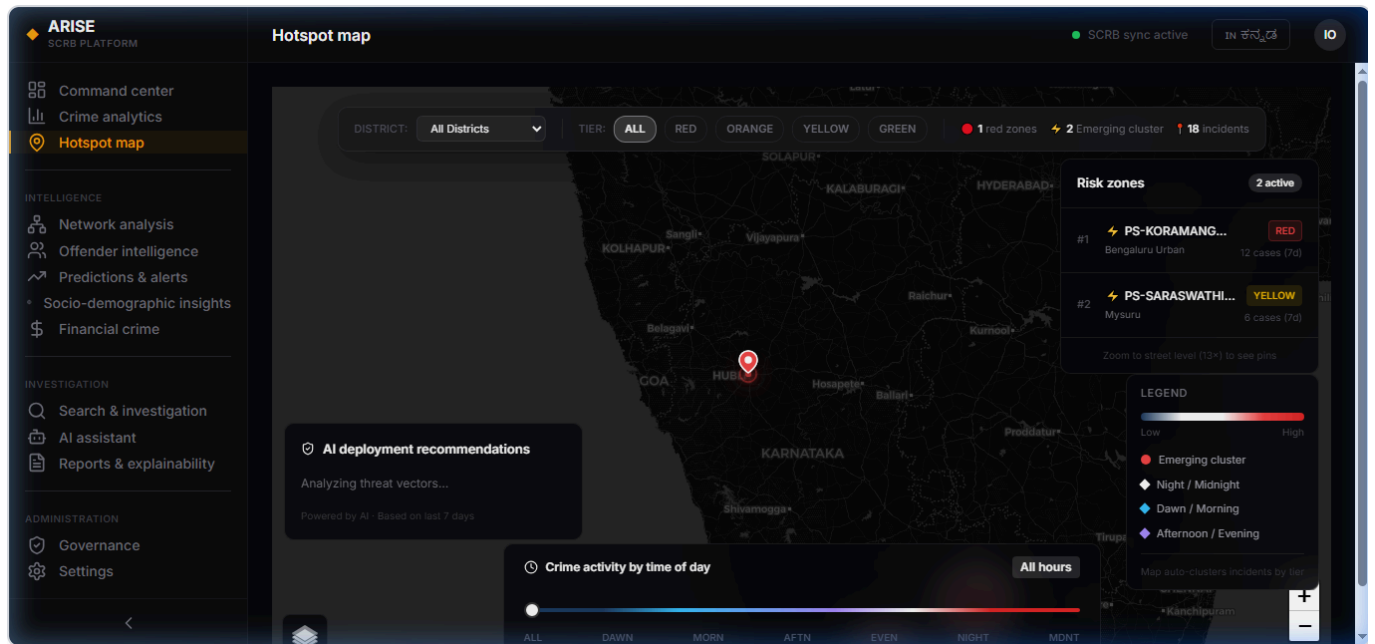


Crime Analytics Screenshot

Section 3: Hotspot Map (`HotspotMap.jsx`)

A geospatial dashboard representing real-time incidents and emerging crime hotspots across Karnataka.

- **Core Features:**
 - Spatiotemporal sliding timeline controller filtering incident layers (Dawn, Morning, Afternoon, Evening, Night, Midnight, All).
 - Drill-down filters by District and Risk Tier (RED, ORANGE, YELLOW, GREEN).
 - **AI Resource Deployment Recommendations Widget:** Recommends patrol frequencies and specific deployment hours based on spatial threat vectors.
- **Visualizations Used:**
 - **Leaflet.js Map with React-Leaflet:**
 - **Heatmap Layer (`leaflet.heat`):** Visual density gradients indicating local risk concentration.
 - **Pulsing Custom Markers:** Custom SVG radar-pulsing rings representing "Emerging Clusters" where crime volume is rapidly surging compared to historical averages.
 - **Pin Layers:** Custom icon markers representing detailed FIR locations with popups.
 - **AI Integration:**
 - **Emerging Cluster Detector:** Identifies and highlights zones experiencing >300% surge in recent crime rates relative to the 30-day baseline.
 - **Geospatial Predictive Recommendation Engine:** Predicts optimal patrol shifts.

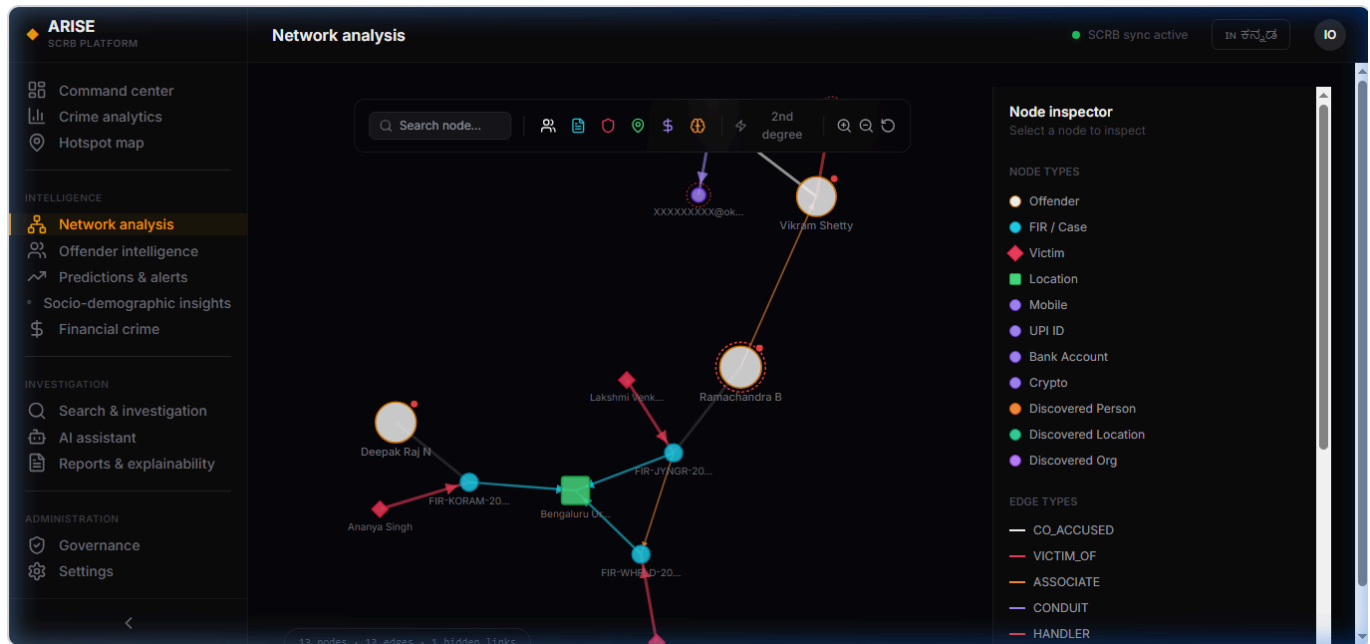


Hotspot Map Screenshot

Section 4: Network Analysis (`NetworkAnalysis.jsx`)

A network visualization mapping relationships between offenders, cases, vehicles, communications, and financial channels.

- **Core Features:**
 - Drill-down investigation graph centered on a selected case or offender.
 - Visual representation of links such as "Co-accused," "Conduit," "Handler," "Family," and "Location Of."
 - Offender risk gauge showing recidivism risk directly in the inspector sidebar.
- **Visualizations Used:**
 - **D3.js Force-Directed Graph:** Fully interactive canvas with custom node symbols (Offenders = circles, FIRs = circles, Victims = diamonds, Locations = squares, Mobile/UPI/Bank/Crypto = specific icons). Nodes are decorated with dashed red lines (Flagged Financials) or dotted borders (Rowdy Sheeters).
- **AI Integration:**
 - **Zia Named Entity Recognition (NER):** Custom nodes labeled as "Zia Discovered" are generated by running NER models over unstructured FIR narratives to extract hidden suspects, mobile numbers, and bank accounts. Showcased with a sparkle (✨) icon and AI extraction confidence score.
 - **Hidden Link Inference:** AI-discovered 2nd-degree paths mapping implicit links between disconnected suspects who share common handlers or operational locations.

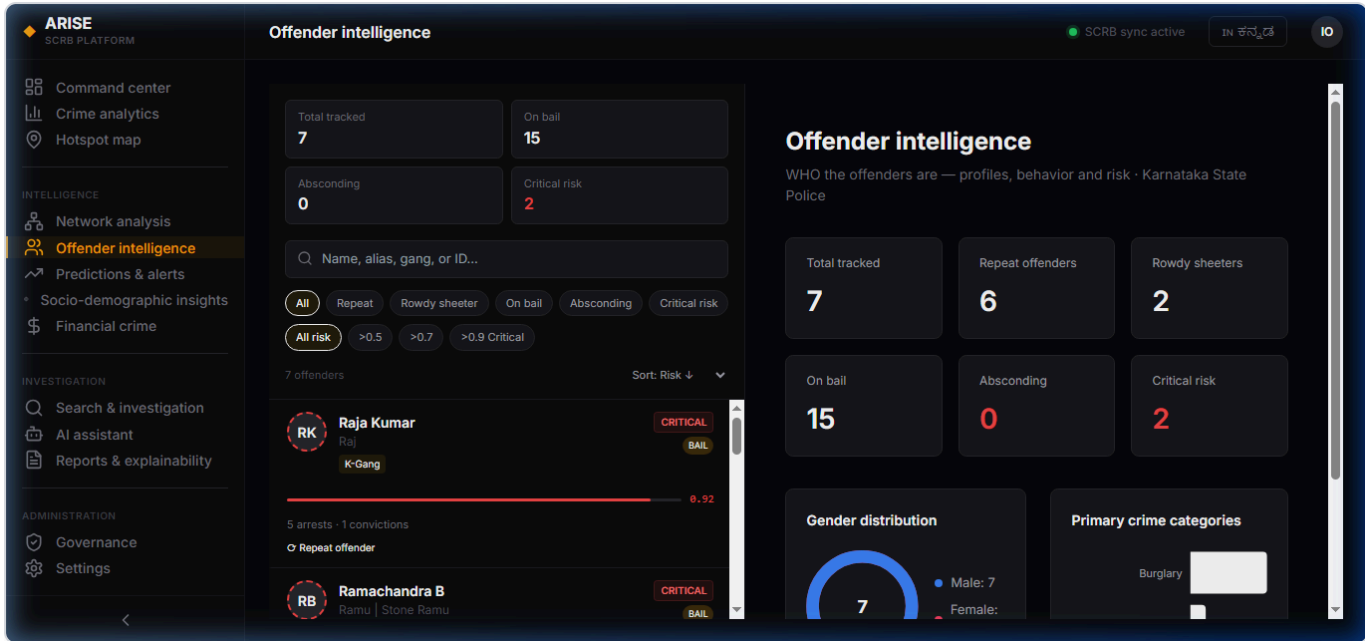


Network Analysis Screenshot

Section 5: Offender Intelligence (offenderIntelligence.jsx)

A profiling interface tracking repeat offenders, history, behavioral traits, and custody status.

- **Core Features:**
 - Tracked offenders database with sorting (Risk score, Prior Arrests, Convictions).
 - Offender behavioral signature profiling outlining preferred operations, time slots, and cross-jurisdictional complexity.
 - Biometric verification panel using facial analytics.
- **Visualizations Used:**
 - **Recidivism Gauge Chart:** Custom semi-circular progress bar displaying the recidivism risk probability.
 - **Recharts Radar Chart:** Five-axis behavioral comparison (Violence, Stealth, Planning, Recidivism, Network complexity) compared against state averages.
 - **Recharts Horizontal Bar Chart:** Primary crime category load.
- **AI Integration:**
 - **Zia Face Analytics:** Processes uploaded suspect photos in real-time, performing face matching against database records and returning age, gender, verification status (**VERIFIED** vs. **DISCREPANCY**), and extraction confidence.
 - **Escalation Pattern Predictor:** Triggers critical warnings if an offender's profile shows a transition towards more violent crime instruments.

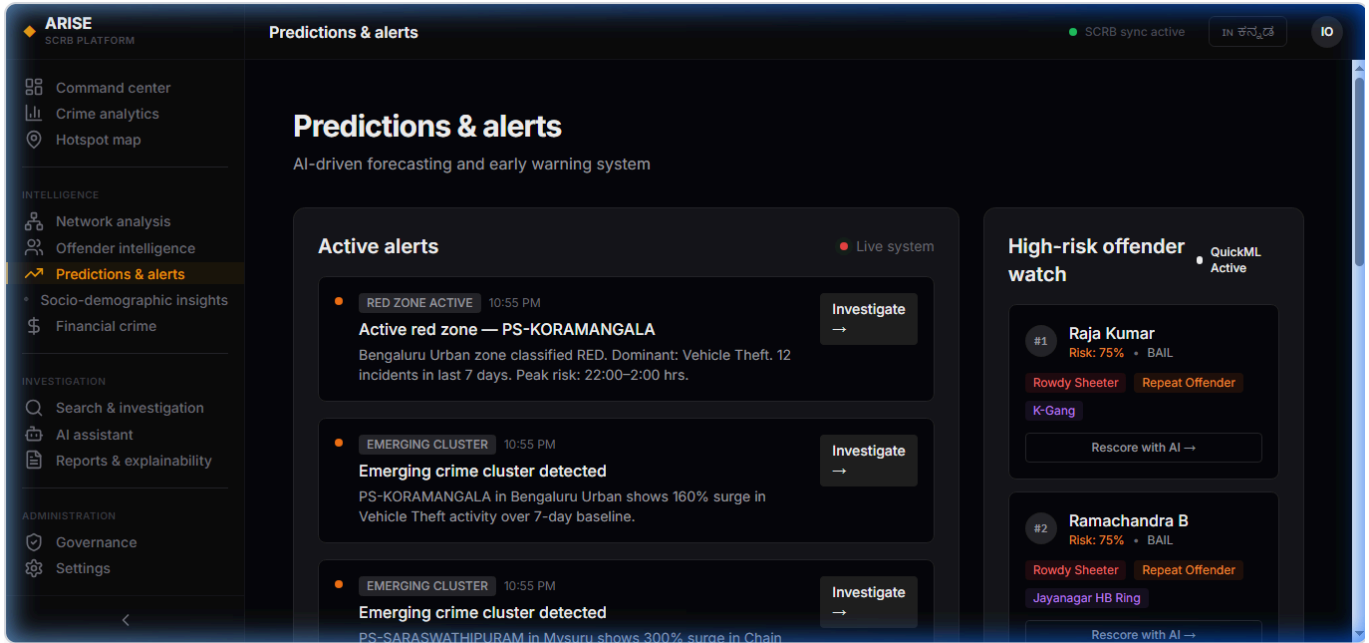


Offender Intelligence Screenshot

Section 6: Predictions & Alerts (Predictions.jsx)

An early-warning forecasting module predicting risk trends and anomalies.

- **Core Features:**
 - Historical vs. predicted crime trend forecasting.
 - High-risk offender leaderboards.
 - Real-time system-wide anomaly detection stream.
- **Visualizations Used:**
 - **Recharts Composed Chart (Line + Area):** Plots historical case numbers alongside AI-projected crime trajectories, with shaded confidence intervals.
 - **Gauge Charts:** Individual risk score representations.
- **AI Integration:**
 - **QuickML Crime Forecasting:** Uses predictive ML models to forecast upcoming regional crime load.
 - **Risk Scoring Leaderboard:** Rank-orders offenders based on calculated recidivism threat scores.
 - **Statistical Anomaly Detection:** Flags statistical irregularities (e.g., sudden spikes in night theft) in specific subdivisions.

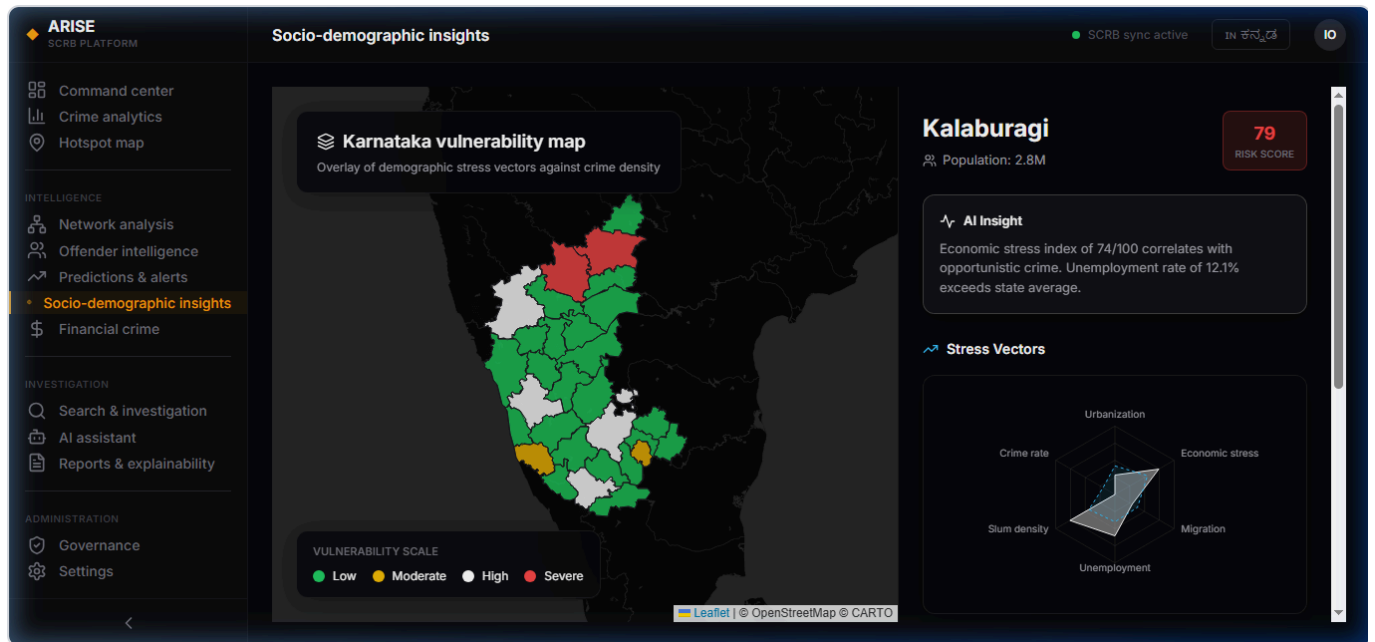


Predictions and Alerts Screenshot

Section 7: Socio-Economic Vulnerability (SocioEconomic.jsx)

Integrates demographic stress factors (economic vulnerability, migration patterns, and unemployment) to analyze root causes of localized crime trends.

- **Core Features:**
 - Choropleth mapping of Karnataka's districts.
 - District-level comparative analysis against state averages.
 - AI-generated district intelligence summaries.
- **Visualizations Used:**
 - **Leaflet.js Choropleth Layer:** Renders dynamic GeoJSON district polygons colored according to their calculated Socio-Economic Vulnerability Index (Green = Low, Yellow = Moderate, Orange = High, Red = Severe).
 - **Recharts Radar Chart:** Overlays district stress vectors (Unemployment proxy, economic stress, migration index, crime rate) against state-wide benchmarks.
- **AI Integration:**
 - **Socio-Demographic AI Insight Engine:** Synthesizes demographic stress parameters to generate natural language explanations explaining **why** a particular district exhibits high crime vulnerability.

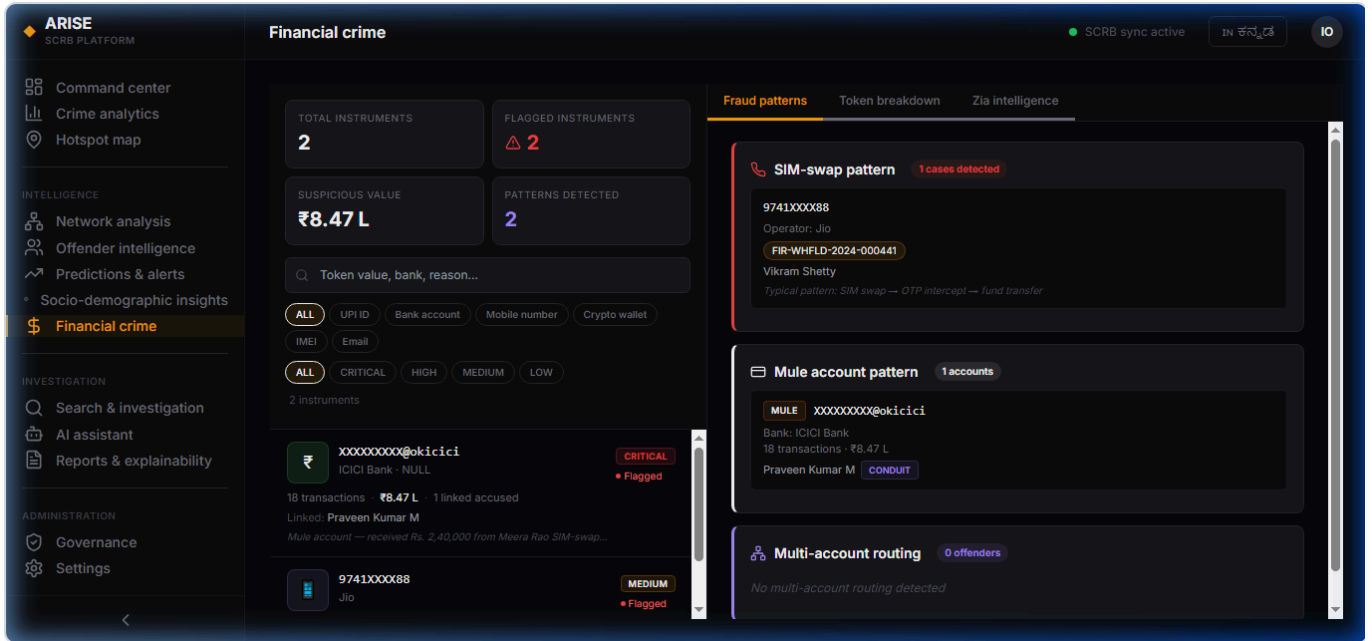


Socio-Economic Vulnerability Screenshot

Section 8: Financial Crime Intelligence (`FinancialCrime.jsx`)

Detects and tracks cyber fraud vectors, suspicious UPI transfers, mule accounts, and shell transactions.

- **Core Features:**
 - Filters for tracking UPI IDs, Bank Accounts, Mobile Numbers, Crypto Wallets, and IMEI.
 - Money Trail Visualizer tracing illicit funds movement across actors.
 - Fraud pattern detection modules (SIM-swap intercepts, mule networks, multi-account routing).
- **Visualizations Used:**
 - **Money Trail Node Map:** A horizontal flow timeline chaining actors, financial instruments, and destination cases.
 - **Distribution Bar Progress Tracks:** Breakdown of flagged instruments by type.
- **AI Integration:**
 - **Zia Text Analytics:** Automatically extracts keywords and suspicious flags from transaction remarks and account reason codes to label fraud patterns.
 - **Fraud Pattern Detection Engine:** Identifies SIM-Swapping networks (multiple numbers mapped to single IMEI), mule accounts, and complex split-routing transactions.

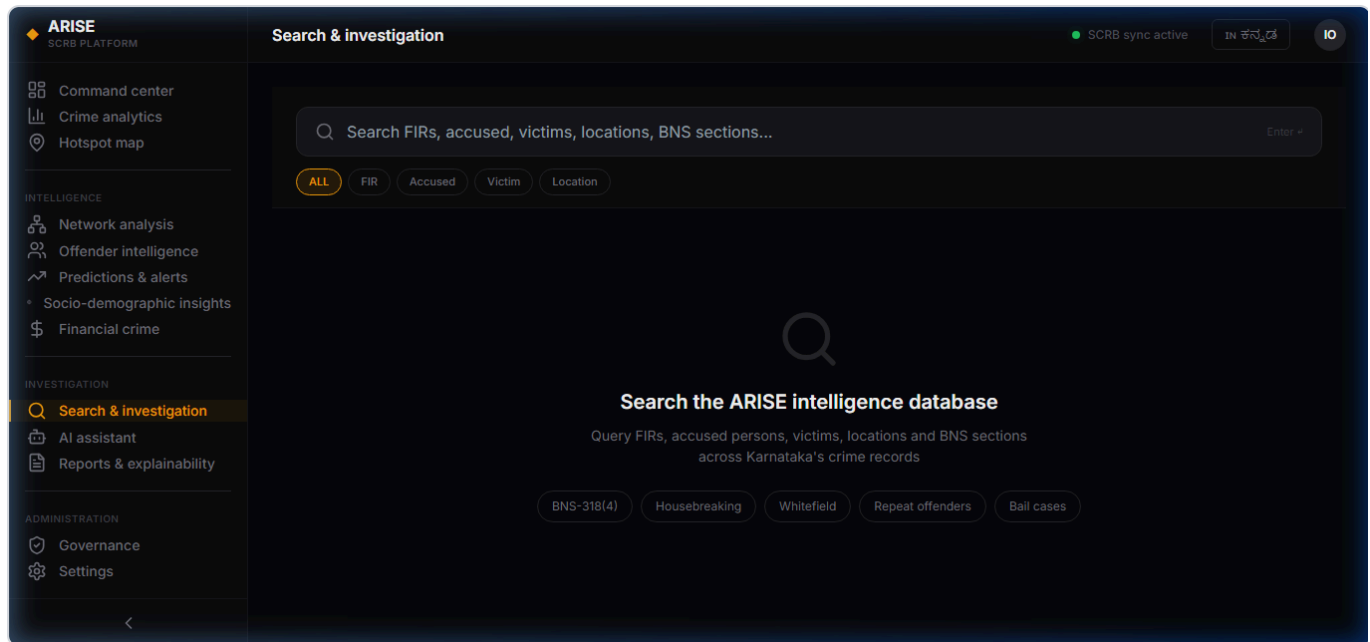


Financial Crime Intelligence Screenshot

Section 9: Search & Investigation (Search.jsx)

A unified search index allowing investigators to run complex queries across cases, offenders, locations, and extracted text.

- **Core Features:**
 - Auto-suggestions dropdown displaying matches as user types.
 - Faceted filtering by BNS Section, District, Crime Category, Status, and Weapons.
 - Interactive preview modals for cases and offender profiles.
- **Visualizations Used:**
 - Clean list structures and preview cards with priority badges.
- **AI Integration:**
 - **Semantic Suggestion System:** Auto-suggests linked records and associated entities based on text matches.

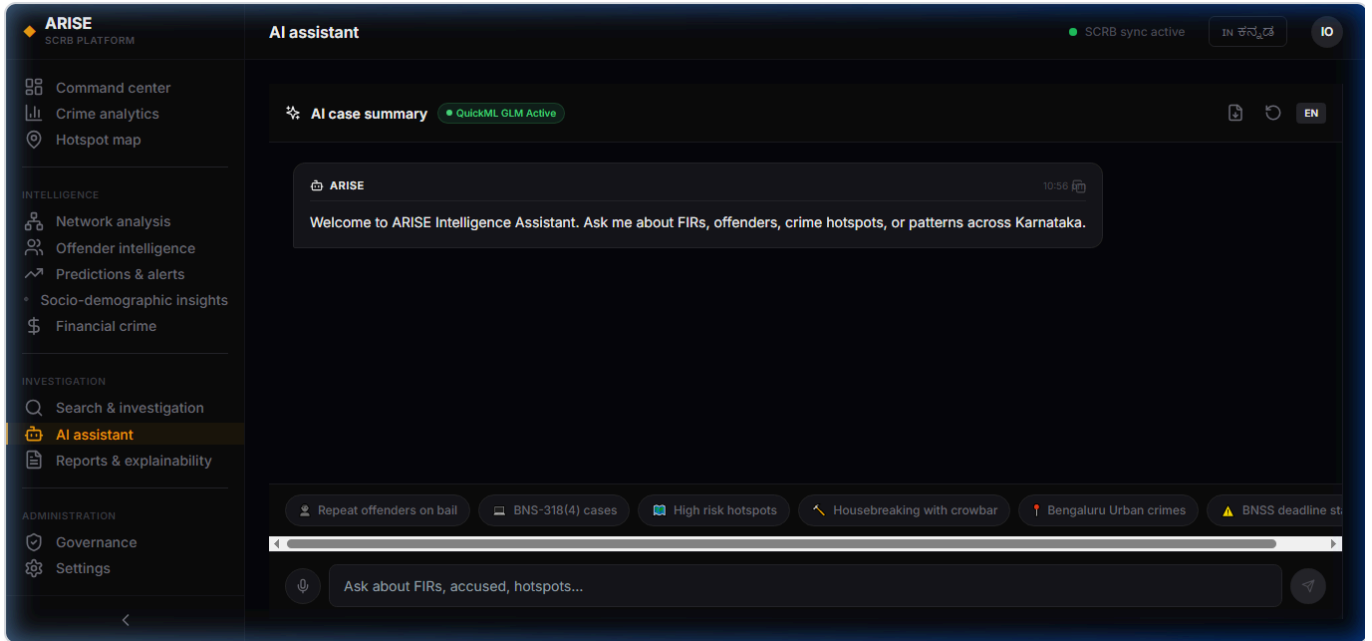


Search and Investigation Screenshot

Section 10: AI Assistant (`AIAssistant.jsx`)

A RAG-powered (Retrieval-Augmented Generation) copilot for field investigators.

- **Core Features:**
 - Dual translation/locale support (English / Kannada) for querying intelligence.
 - Voice Querying (Speech-to-Text) and Text-to-Speech audio playbacks.
- **RAG Citations and Source-Doc Verification:** Displays source documents (e.g., Table name, Row UID) for every generated fact, preventing hallucinations.
- **AI Integration:**
 - **Catalyst GLM (Generative Large Language Model):** Powers conversational QA over case histories.
 - **Zia Speech Synthesis (TTS) & Recognition (STT):** Handles native multilingual voice controls.
 - **ConvoKraft Chatbot Handler:** Orchestrates ZCQL queries under the hood using intent classification.

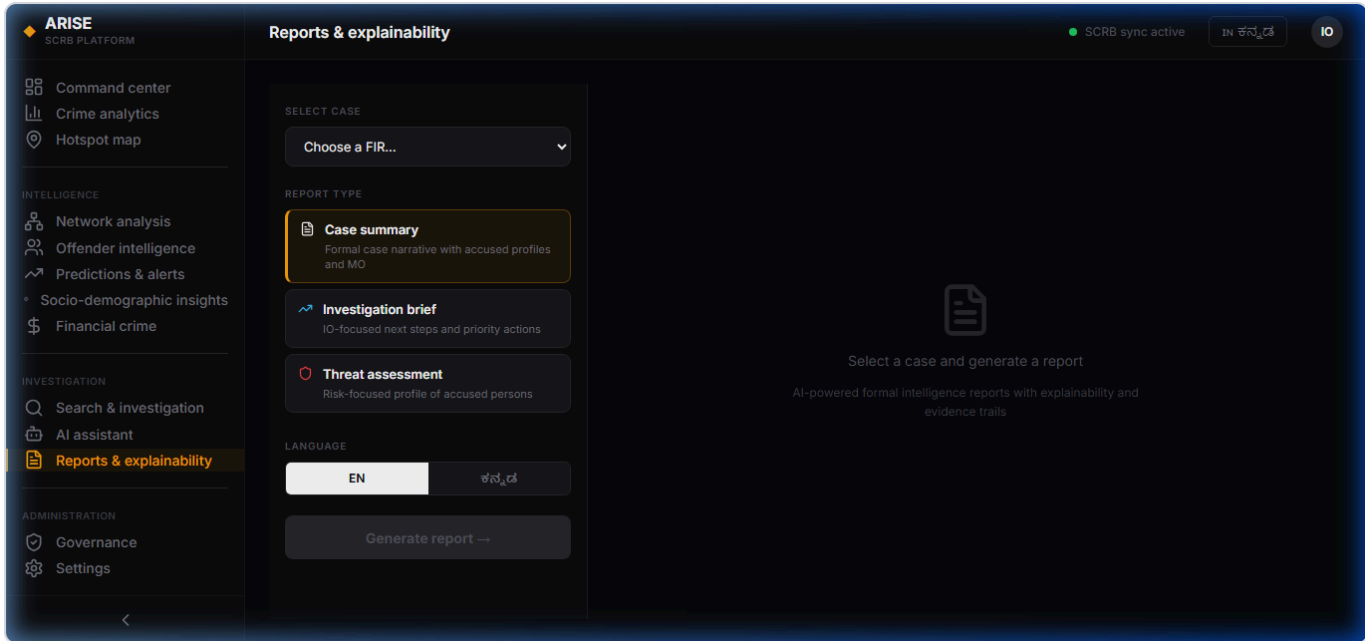


AI Assistant Screenshot

Section 11: Reports & Explainability ([Reports.jsx](#))

Formulates legally compliant, audit-secure formal case summaries.

- **Core Features:**
 - Selectable output types: Case Summary, Investigation Briefs, and Threat Assessments.
 - **BSA Sec. 63 Compliance Evidence Trail:** Creates a tamper-evident audit timeline of record access events.
 - Similar Cases Finder.
- **Visualizations Used:**
 - **Recharts / Custom Gauge SVG:** Visual representation of AI reasoning confidence.
 - **Audit Log Timeline:** Visual timeline of access records with color-coded action pills (INSERT, UPDATE, SELECT, DELETE).
- **AI Integration:**
 - **Report Generation Engine:** Generates localized case narratives using Catalyst GLM.
 - **Explainability Factor Model:** Details **why** the AI reached a given conclusion, complete with weighting percentages and data source citations.
 - **Tamper Detection Engine:** Flagging system that highlights unauthorized modifications or anomalous data interactions.



Reports and Explainability Screenshot

3. Technology Stack & Architecture Summary

- **Frontend Core:** React, Vite, Lucide React (Icons).
- **Visualizations:** D3.js (Force graphs), Leaflet / React-Leaflet (Geospatial and Choropleths), Recharts (Analytical charts).
- **Backend & Hosting:** Zoho Catalyst Serverless platform.
- **Datastore Engine:** Catalyst DataStore queried via ZCQL.
- **AI Engine Stack:**
 - **Zia Services:** Face Analytics, Speech STT/TTS, Text Analytics.
 - **QuickML:** Predictive modeling and anomaly detection.
 - **Catalyst GLM:** Generative AI, RAG chat, and report writer.
- **Security & Governance:** BSA Sec. 63 compliant SHA-256 integrity hash chains.